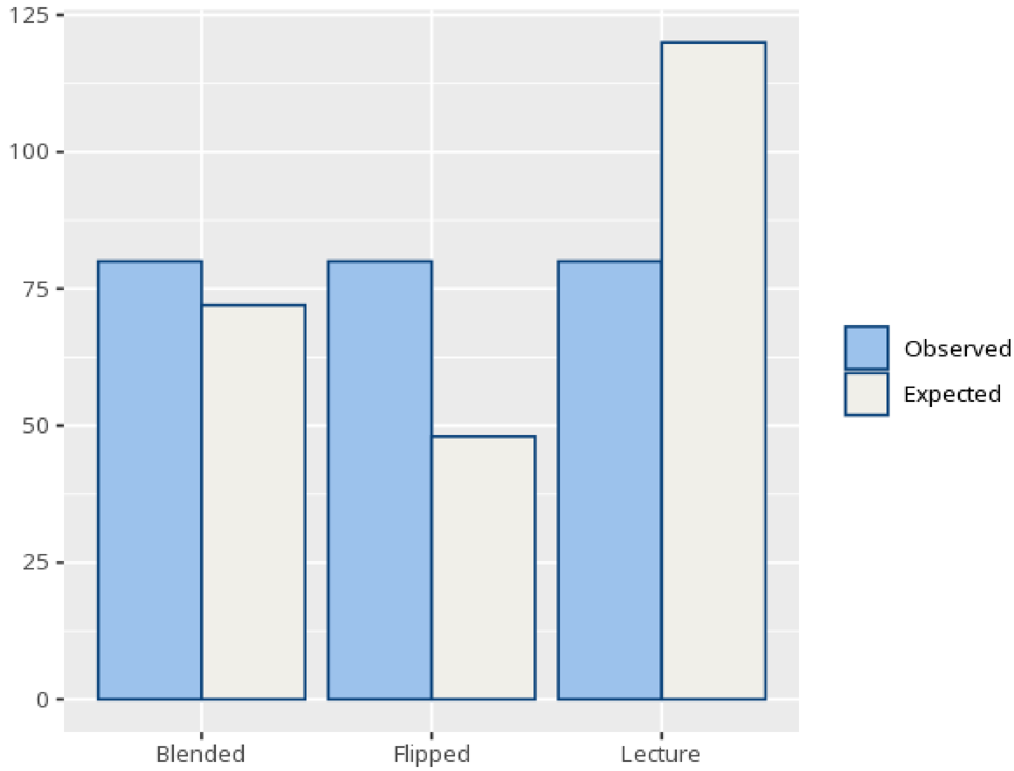
1 Chi-square goodness-of-fit

Category	Observed	Expected	Std. residual
Blended	80	72.00	1.13
Flipped	80	48.00	5.16
Lecture	80	120.00	−5.16

χ^2	df (k−1)	p	Cohen's w [95% CI]
35.56	2	<.001	0.38 [0.27, 2.00]



Tests whether one categorical variable's counts depart from an expected distribution (equal by default). A p below alpha means the observed split differs from expected; a standardized residual beyond about ±1.96 flags a category that significantly drives the result. Valid only when expected counts are mostly ≥ 5 (some texts allow ≥ 1 if no more than 20% are below 5); for sparse categories use an exact / simulated test instead. Your significance threshold (α) is 0.05.

A goodness-of-fit test, $\chi^2(2, N=240)=35.56$, $p < .001$, w=.38 [.27, 2.00].

Statistical basis: Chi-square goodness-of-fit test (Pearson, K. (1900). "On the criterion that a given system of deviations from the probable in the case of a correlated system of variables is such that it can be reasonably supposed to have arisen from random sampling." Philosophical Magazine, 50(302), 157-175.);

Cohen's w effect-size benchmark (Cohen, J. (1988). Statistical Power Analysis for the Behavioral Sciences (2nd ed.). Routledge.).